

Differentiability

Differentiability

A function $f(x)$ is said to be differentiable at a point $x = a$ if

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \text{ exist.}$$

Again $f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$ will exist if

$$\lim_{h \rightarrow 0} \frac{f(a-h) - f(a)}{-h} \quad \text{and} \quad \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

In this case, $\lim_{h \rightarrow 0} \frac{f(a-h) - f(a)}{-h}$ is called Left Hand Derivative $[L f'(a)]$.

and $\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$ is called Right Hand Derivative $[R f'(a)]$.

Problem

Show that $f(x)$ is continuous but not differentiable at $x=1$ and $x=2$.

Solution

$$f(x) = \begin{cases} -3x + 5; & -3 < x < 1 \\ 2; & 1 \leq x < 2 \\ 1 + \frac{1}{2}x; & 2 \leq x < 4 \end{cases}$$

When $x \leq 1$ **then** $f(x) = -3x + 5$

Now L.H.D $= Lf'(1)$

$$= \lim_{h \rightarrow 0} \frac{f(1-h) - f(1)}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{\{-3(1-h) + 5\} - 2}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{\{-3 + 3h\} + 5 - 2}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{3h}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{3}{-1} = -3$$

When $x > 1$ **then** $f(x) = 2$

Now R.H.D $= Rf'(1)$

$$= \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2 - 2}{h}$$

$$= 0$$

Since $Lf'(1) \neq Rf'(1)$

The function $f(x)$ is not differentiable at $x = 1$.

Try yourself to test differentiability at $x = 2$.

Problem Show that $f(x)$ is continuous but not differentiable at $x=0$.

Solution

$$f(x) = \begin{cases} 3 + 2x, & -\frac{3}{2} \leq x \leq 0 \\ 3 - 2x, & 0 < x < \frac{3}{2} \end{cases}$$

When $x \leq 0$ then $f(x) = 3 + 2x$

Now L.H.D $= Lf'(0)$

$$\begin{aligned} &= \lim_{h \rightarrow 0} \frac{f(0-h) - f(0)}{-h} \\ &= \lim_{h \rightarrow 0} \frac{f(-h) - f(0)}{-h} \\ &= \lim_{h \rightarrow 0} \frac{(3 - 2h) - (3 + 2 \cdot 0)}{-h} \\ &= \lim_{h \rightarrow 0} \frac{-2h}{-h} = 2 \end{aligned}$$

When $x > 0$ then $f(x) = 3 - 2x$

Now R.H.D $= Rf'(0)$

$$\begin{aligned} &= \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(3 - 2h) - (3 + 2 \cdot 0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{-2h}{h} = -2 \end{aligned}$$

Since $Lf'(0) \neq Rf'(0)$

The function $f(x)$ is not differentiable at $x = 0$.

Problem

Show that $f(x)$ is continuous and differentiable at $x = 0$

$$f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right), & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$$

Solution

Now L.H.D = $Lf'(0)$

$$= \lim_{h \rightarrow 0} \frac{f(0-h) - f(0)}{-h}$$

$$= \lim_{h \rightarrow 0} \frac{(-h)^2 \sin\left(-\frac{1}{h}\right) - 0}{-h}$$

$$= \lim_{h \rightarrow 0} h \sin \frac{1}{h}$$

$$= 0 \quad [\text{Using Sandwich theorem}]$$

$$\left[\begin{array}{l} \text{We know, } -1 \leq \sin \frac{1}{h} \leq 1 \\ \lim_{h \rightarrow 0} -h \leq \lim_{h \rightarrow 0} h \sin \frac{1}{h} \leq \lim_{h \rightarrow 0} h \\ 0 \leq \lim_{h \rightarrow 0} h \sin \frac{1}{h} \leq 0 \end{array} \right]$$

Now R.H.D = $Rf'(0)$

$$= \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h^2 \sin \frac{1}{h} - 0}{h}$$

$$= \lim_{h \rightarrow 0} h \sin \frac{1}{h}$$

$$= 0$$

Since $Lf'(0) = Rf'(0)$

The function $f(x)$ is differentiable at $x = 0$.

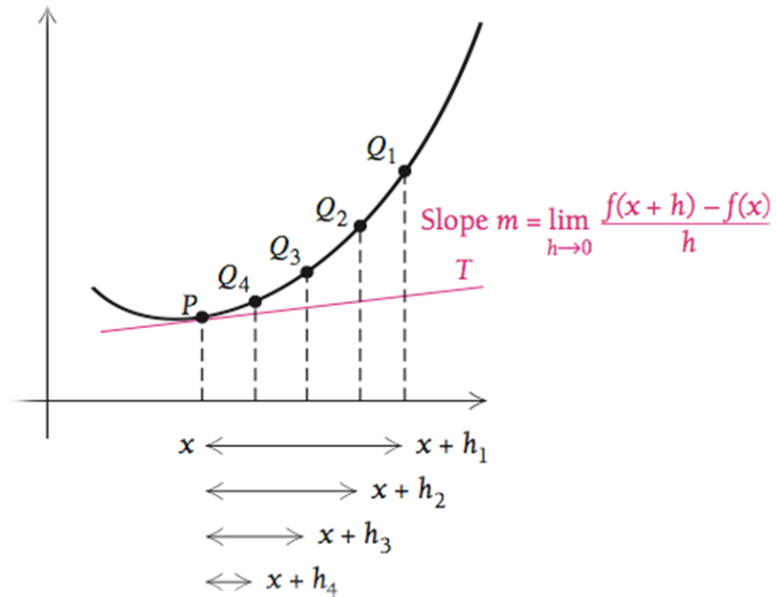
Differentiation Using Limits of Difference Quotients

DEFINITION:

The **slope of the tangent line** at $(x, f(x))$ is

$$m = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

This limit is also the **instantaneous rate of change** of $f(x)$ at x .



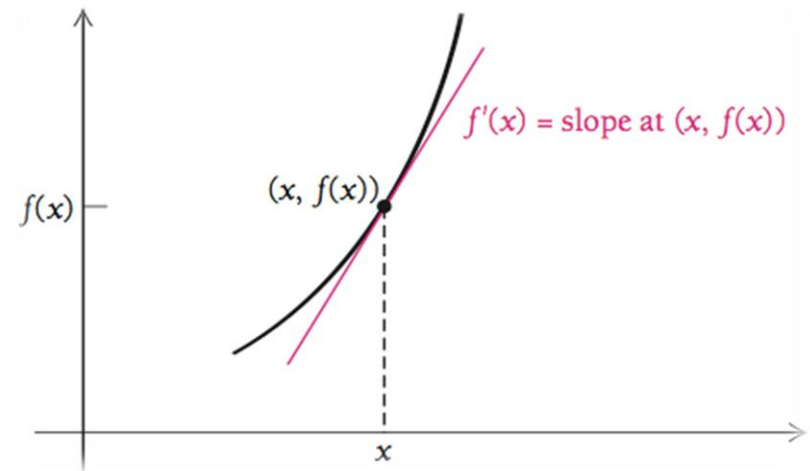
DEFINITION:

For a function $y = f(x)$,
its **derivative** at x is the
function defined by

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

provided the limit exists.

If $f'(x)$ exists, then we say
that f is **differentiable** at x .



Example: For $f(x) = x^2$, find $f'(x)$. Then find $f'(-3)$ and $f'(4)$.

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{\cancel{x^2} + 2xh + h^2 - \cancel{x^2}}{h} = \lim_{h \rightarrow 0} \frac{\cancel{h}(2x+h)}{\cancel{h}}$$

$$f'(x) = \lim_{h \rightarrow 0} 2x + h$$

$$f'(x) = 2x$$

$$f'(-3) = 2(-3) = -6$$

$$f'(4) = 2(4) = 8$$

Example : For $f(x) = x^3$, find $f'(x)$.

Then find $f'(-1)$ and $f'(1.5)$.

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{(x+h)^3 - x^3}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{\cancel{x^3} + 3x^2h + 3xh^2 + h^3 - \cancel{x^3}}{h}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{\cancel{h}(3x^2 + 3xh + h^2)}{\cancel{h}} = \lim_{h \rightarrow 0} 3x^2 + 3xh + h^2$$

$$f'(x) = 3x^2$$

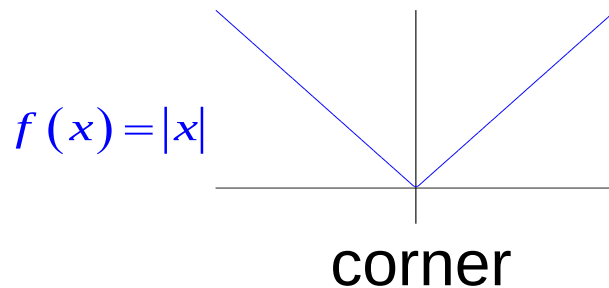
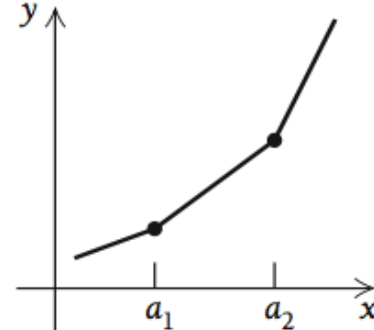
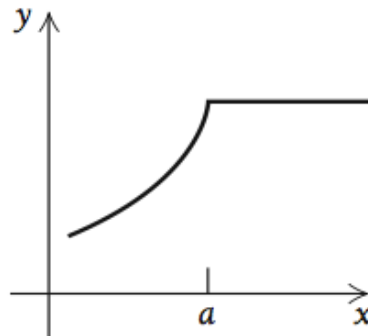
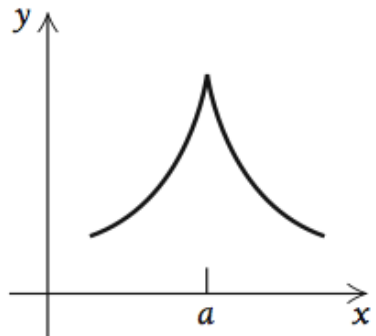
$$f'(-1) = 3(-1)^2 = 3(1) = 3$$

$$f'(1.5) = 3(1.5)^2 = 3(2.25) = 6.75$$

Where a Function is Not Differentiable:

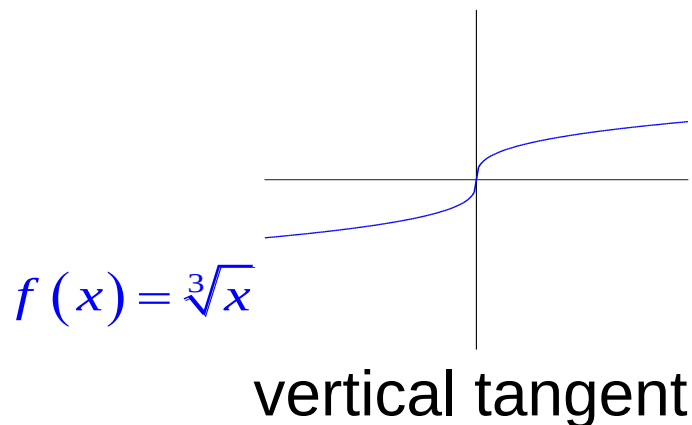
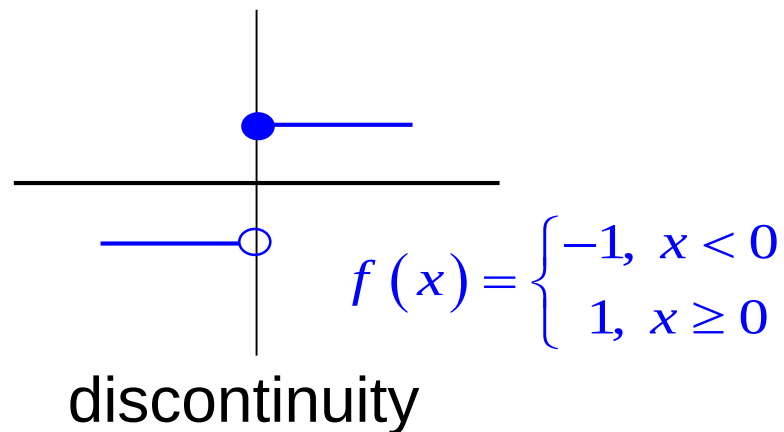
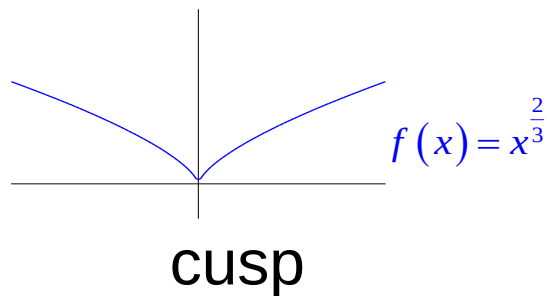
To be differentiable, a function must be continuous and smooth.

Derivatives will fail to exist at:



- **A function $f(x)$ is not differentiable at a point $x = a$, if there is a “corner” at a .**

Derivatives will fail to exist at:



- A function $f(x)$ is not differentiable at a point $x = a$,
if there is a **cusp**, **discontinuity** and **vertical tangent** at a .

Thank you